Data Analysis

BY THE END:

You can explain how data analysis turns raw data into findings that support decisions.

Overview

A shop owner notices that one product sells out every week while another sits on the shelf. She wants to understand why and decide what to order next. This is where **data analysis** becomes useful.

What is data analysis?

Data analysis is the process of inspecting, cleaning, transforming, and interpreting data to discover useful information and support conclusions.

Data analysis may involve:

- Exploring sales trends
- Comparing performance across regions
- Identifying reasons for customer churn
- Preparing reports for managers

Data analysis often answers questions like:

Which products sold the most?

Why did revenue drop in March?

Which customer group is most profitable?

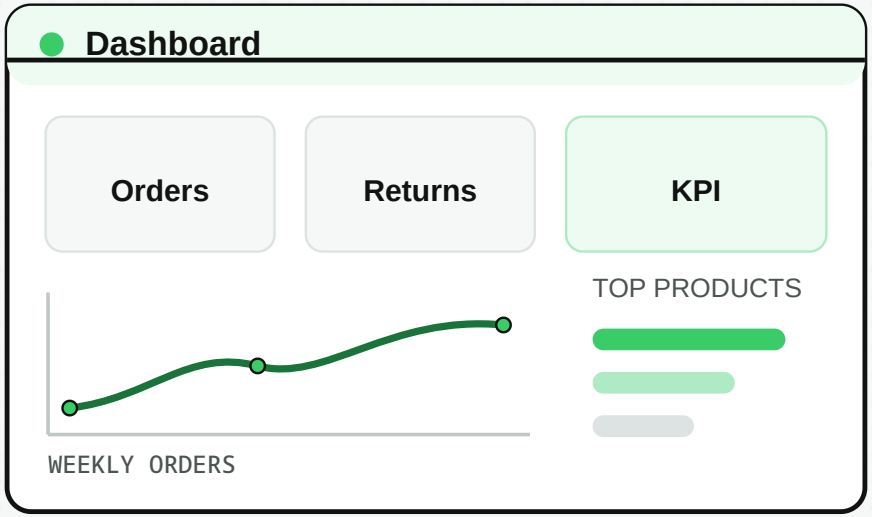
What data analysis can produce

The result depends on the question. Data analysis can produce:

- A finding that explains what happened
- A report that communicates the evidence
- A metric that tracks something important
- A dashboard that keeps results visible over time

A KPI (key performance indicator) is a metric used to measure progress toward a goal.

Regular dashboards and performance reports are often called **business intelligence (BI)**. Companies may use different names for this work, but the purpose is the same: make useful information available for decisions.



Example

A wholesaler supplies shops in Springfield, Riverton, and Lakeview, and wants to know how each is doing.

- The analysis compares six months of orders and returns
- It finds that the Lakeview shops order the same volume as the others but return far more unsold stock
- The result is shared in a report and a dashboard showing each city's weekly orders, returns, and top products

The analysis answered the manager's question and gave her evidence for the next ordering decision.

